☁ CLOUD + ⬡ EDGE — CLOUD-AGNOSTIC AI ARCHITECTURE

JETSON ORIN · QUALCOMM RB5 · RASPBERRY PI 5 · FLEET 2–10 · OPC-UA · MODBUS · SQLITE · MLFLOW

✓ AWS ✓ Azure ✓ Google Cloud ✓ Self-hosted / On-premise — select any cloud, edge device is identical —

Design principle: Every cloud component maps to an equivalent on AWS, Azure, and GCP — shown below each service. The edge device stack is **100% cloud-agnostic**: Docker + FastAPI + Mosquitto + SQLite run identically on Jetson, Qualcomm, and RPi regardless of which cloud is chosen. Switch cloud providers without touching a single line of edge code.

☁ CLOUD — MLOPS PIPELINE (CHOOSE YOUR CLOUD OR SELF-HOST)

① DATA INGESTION · STORAGE · LABELING

☁ Object Storage: AWS S3 | Azure Blob Storage | GCS | *MinIO (self-hosted, S3-compatible)*

Data Ingestion Service
Upload images · video · sensor CSV from edge or browser to object storage
FastAPI upload endpoint | boto3 / azure-storage / GCS client | DVC (dataset versioning) | rclone (cloud-agnostic sync)
Edge devices auto-upload anomaly frames for retraining via HTTPS + API key

Image Labeling
Bounding boxes · segmentation · classification — all open-source, cloud-agnostic
Label Studio (Docker, any cloud) | CVAT (Docker, any cloud) | Roboflow (SaaS, cloud-neutral) | FiftyOne (local/cloud)

Dataset Registry
Versioned train / val / test splits · lineage · reproducibility
DVC + object storage remote | HuggingFace Datasets | Roboflow Universe | Git + Git LFS

② MODEL ZOO — PULL FROM ANY REGISTRY (SOURCE IS ALWAYS CLOUD-AGNOSTIC)

GitHub
YOLOv8/11 · RT-DETR · custom architectures
git clone | pip install

HuggingFace Hub
ViT · DETR · CLIP · LLMs · vision models
transformers | timm

Qualcomm AI Hub
Snapdragon-optimised models
qai_hub SDK | SNPE / QNN

NVIDIA NGC
TAO · PeopleNet · DetectNet
ngc pull | TAO Toolkit

ONNX Model Zoo
Universal format · hardware-neutral
onnx model zoo | timm export

③ FINE-TUNING · TRAINING · EXPERIMENT TRACKING — MLFLOW ★ RECOMMENDED: OPEN-SOURCE, CLOUD-AGNOSTIC

☁ GPU Training VM: AWS EC2 p3/p4 (spot) | Azure NC / NDv4 (spot) | GCP A100 / T4 (preemptible) | *RunPod · Lambda Labs · Vast.ai (GPU cloud)*

Training Pipeline — Cloud GPU VM
Fine-tune on custom labelled data · GPU spot / preemptible instances · distributed training
PyTorch | Ultralytics YOLO | HuggingFace Trainer | PyTorch Lightning | Kubeflow Pipelines (k8s-native) | GitHub Actions (CI trigger) | GitLab CI / Jenkins
Same PyTorch training code runs on any cloud GPU — no cloud SDK needed in training loop

MLflow — Experiment Tracking + Model Registry ★ Free · self-hosted · cloud-agnostic
Why MLflow over cloud-native: same API on AWS, Azure, GCP, local — zero vendor lock-in.
Run on a single VM (any cloud) · S3-compatible artifact store · PostgreSQL backend
MLflow Tracking UI :5000 | mAP · loss · confusion matrix · hyperparams | MLflow Model Registry (Staging→Production) | TensorBoard (embedded) | DVC (dataset + model versioning)
Alternatives: Weights & Biases (SaaS, cloud-neutral), ClearML (open-source), Neptune.ai

④ MODEL OPTIMIZATION (PER TARGET) · VALIDATION GATE · DOCKER BUILD · CONTAINER REGISTRY

☁ Container Registry: AWS ECR | Azure Container Registry | Google Artifact Registry | *GHCR · Docker Hub · Harbor (self-hosted)*

⚙ Optimize Per Target Hardware
One .pt → 3 optimised artifacts from same training output
TensorRT → .engine (Jetson) | SNPE / QNN → .dlc (Qualcomm) | TFLite → .tflite (RPi 5) | ONNX Runtime (universal fallback) | Qualcomm AI Hub compile API
Optimization tools are vendor-specific (NVIDIA/Qualcomm) but cloud-agnostic — run on any GPU VM

✅ Validation Gate (CI/CD)
Must pass before Docker image is pushed to registry
pytest — accuracy ≥ baseline | latency benchmark on sim device | GitHub Actions (cloud-agnostic) | GitLab CI / Jenkins / Tekton | Docker image signing (cosign) | SHA-256 model checksum
Never push untested model to production device fleet

Docker Build (multi-arch ARM64)
Separate image per microservice · model weights NOT in image
Docker Buildx (multi-arch) | Multi-stage Dockerfile | NVIDIA L4T base (Jetson) | Alpine / Debian slim (RPi) | GitHub Actions / GitLab CI

Container Registry (OCI-standard)
Versioned + signed images per service per device target
Any OCI-compatible registry | GHCR (GitHub, free) | Harbor (self-hosted) | :inference-v3-jetson-arm64 | :inference-v3-qualcomm-arm64 | :inference-v3-rpi-arm64

⑤ OTA FLEET DELIVERY — 2–10 DEVICES · DOCKER-COMPOSE PUSH · MODEL WEIGHTS PUSH (CLOUD-AGNOSTIC)

☁ Device Management: AWS IoT Greengrass v2 | Azure IoT Edge | GCP IoT Core + Edge TPU | *Balena (open-source, any cloud) · Ansible + SSH · Portainer*

OTA Device Management ★ Balena recommended
Cloud-agnostic fleet management · container OTA · remote shell
Balena Cloud (cloud-neutral) | OpenBalena (self-hosted) | Watchtower (per-device, any registry) | Ansible playbooks (SSH fleet) | Portainer (container UI)
Balena: works with any cloud, any registry, any device. One dashboard for 2–10 devices. Free tier available.

⬇ Model Weights Push (independent of image)
Code fix = small image pull. Model update = weights only, no image rebuild.
rsync over SSH / SCP | SHA-256 checksum verify | update_model.sh script | hot-reload /reload_model API | shadow deploy → promote

Docker Compose + Config Push
Per-device compose file + environment vars
docker-compose.yml | .env (per-device variables) | Ansible (push to fleet) | Balena fleet config | Object storage config bucket

☁ Cloud Data + Remote Fleet UI
Fleet telemetry · alerts · training data store
PostgreSQL (any cloud VM) | Grafana (any cloud/self-hosted) | Prometheus + Alertmanager | EMQX / HiveMQ (cloud MQTT broker) | Nginx (reverse proxy)

⬇ Container pull (OCI registry) · model weights (rsync+SHA256) · docker-compose.yml · .env config | ⬆ Inference logs · OPC-UA / Modbus telemetry · alerts · new training images · drift metrics | 🔒 TLS mutual auth · Tailscale VPN (cloud-agnostic) · SSH tunnel · Balena VPN

⬡ EDGE DEVICE — JETSON ORIN · QUALCOMM RB5 · RASPBERRY PI 5 [100% CLOUD-AGNOSTIC]

Fleet 2–10 · Docker Compose · systemd · ARM64 · identical on all cloud configs

⚙ INDUSTRIAL PROTOCOL BUS — OPC-UA (primary) · Modbus RTU (RS-485) · Modbus TCP · CANBUS · GPIO · I2C · SPI · opcua-asyncio · pymodbus · python-can · PySerial

① SENSOR ACQUISITION LAYER — DOCKER CONTAINERS · FASTAPI + UVICORN · NO CLOUD SDK

📷 Camera Service `:8001`
USB · CSI · RTSP IP cam · MJPEG stream · single frame endpoint
OpenCV · GStreamer · FastAPI + Uvicorn · aiortc (WebRTC) · paho-mqtt publish frame_ready

🔌 OPC-UA / Modbus Service `:8005`
OPC-UA node reads · Modbus RTU/TCP poll · CANBUS · PLC alarms
opcua-asyncio · pymodbus · python-can · FastAPI + Uvicorn · paho-mqtt publish
OPC-UA: richer data model, built-in security, standard for modern PLCs and SCADA

🌡 GPIO / I2C Sensor Service `:8006`
Temp · vibration · pressure · IMU · GPIO · I2C sensors
gpiozero · CircuitPython · PySerial · FastAPI + Uvicorn · paho-mqtt publish

MQTT MESSAGE BUS — ON-DEVICE, CLOUD-AGNOSTIC — Mosquitto :1883 · edge/camera/frame_ready · edge/opcua/+ · edge/modbus/+ · edge/sensor/+ · edge/inference/result · edge/threshold/breach · edge/system/health · fastapi-mqtt · paho-mqtt

② PRE-PROCESSING + INFERENCE — MODEL FROM DOCKER VOLUME /MODELS/ — NOT IN IMAGE — NOT CLOUD-DEPENDENT

🔄 Pre-process Service `:8002`
Resize · normalize · denoise · frame→tensor · time-series windowing
OpenCV · NumPy · FastAPI · Pillow · scikit-learn scaler

🧠 CV Inference Service `:8003`
YOLO · DNN · object detect · classify · model from /models/ Docker Volume
TensorRT .engine (Jetson) · SNPE/QNN .dlc (Qualcomm) · TFLite .tflite (RPi 5) · ONNX Runtime (universal) · Ultralytics YOLO · FastAPI + fastapi-mqtt

📈 Time-Series Inference `:8007`
Sensor anomaly · predictive maintenance · LSTM · TCN
ONNX Runtime · TFLite · Edge Impulse SDK · FastAPI · paho-mqtt publish result

③ BUSINESS LOGIC · THRESHOLD · ALERTS · LOCAL STORAGE — SQLITE ★ RECOMMENDED FOR EDGE

🚨 Threshold / Alert Service `:8004`
Confidence gates · count limits · sensor rules · breach log
FastAPI · fastapi-mqtt subscriber · Pydantic rules engine · paho-mqtt publish breach · SMTP / webhook alert · Alertmanager (Prometheus)

🗄 SQLite — Local Storage ★ Zero config, cloud-agnostic
Sensor readings · inference results · breach log · ~500KB RAM · no daemon · 1 file in Docker Volume
SQLite3 · SQLAlchemy (async) · aiosqlite · WAL mode (concurrent reads) · periodic sync → cloud PostgreSQL
Single .db file persists in Docker Volume. Grafana SQLite plugin reads directly. Works completely offline.

☁ Cloud Telemetry Bridge
Forward alerts + metrics · upload training images · generic protocol
MQTT over TLS (to cloud broker) · EMQX / HiveMQ / Mosquitto bridge · HTTPS REST (cloud-agnostic) · Kafka bridge (if needed) · rclone / object storage SDK
Use standard MQTT/HTTPS — swap cloud endpoint in .env without code changes

④ DASHBOARD — LOCAL (ON-DEVICE LAN) + REMOTE (TAILSCALE VPN — CLOUD-AGNOSTIC)

📊 Live CV Dashboard `:8501`
Live MJPEG stream + detection overlay · breach log · FPS · confidence charts
accessible locally AND remotely — no cloud dependency
Streamlit · FastAPI /video MJPEG · fastapi-mqtt subscriber · Plotly charts

📈 Grafana Dashboard `:3000`
Time-series · sensor trends · inference latency · alert history
Grafana OSS · SQLite datasource plugin · Prometheus datasource · Infinity datasource (REST API)

🌐 Remote Access ★ Tailscale
Dashboard + stream remotely · no port-forwarding · works on 4G · any cloud
Tailscale VPN (cloud-agnostic mesh) · Cloudflare Tunnel (zero-trust) · Nginx reverse proxy · WireGuard (self-hosted VPN)
Tailscale: free ≤100 devices · works behind NAT/4G · 5-min setup · no cloud config changes

⑤ DEVICE INFRASTRUCTURE — IDENTICAL STACK ON JETSON, QUALCOMM, AND RPI — NO CLOUD-SPECIFIC DEPENDENCIES

🐳 Docker Compose Orchestration
All microservices · GPU passthrough · auto-restart · OTA pull from any OCI registry
Docker Engine · Docker Compose v2 · NVIDIA Container Toolkit (Jetson) · Watchtower (any registry, cloud-agnostic) · Balena Supervisor (optional)

📦 Docker Volumes (persist across image updates)
model_weights/ → .engine / .dlc / .tflite · sqlite_data/ → inference.db · config/ → config.yaml, labels.txt · grafana_data/ · prometheus_data/ · logs/
Volumes survive image updates. Switch cloud provider — volumes untouched on device.

🔍 Monitoring + Health
GPU · CPU · RAM · FPS · inference latency · service uptime
Prometheus :9090 · Grafana (local, same instance) · jtop (Jetson GPU monitor) · /healthz FastAPI endpoints · systemd watchdog + restart

⚡ OPTIONAL LAYER — AGENTIC / LLM / ROBOTICS (ADD WHEN READY, FULLY ON-DEVICE)

🤖 On-device LLM
Tool-calling orchestrator · Qwen3-4B · Llama3.2-3B
Ollama · llama.cpp · Uvicorn
Jetson Orin Nano 8GB or Qualcomm RB5. No cloud dependency.

🔌 MCP Servers
Sensor · Inference · Dashboard MCP tools
MCP Python SDK · FastAPI · STDIO transport

🦾 ROS2 Bridge (robotics)
Natural language → robot commands · Nav2 · SLAM
ROS2 Humble · llama_ros · Gazebo

🧠 Agent Decision Loop
sensor → LLM → action · 3–10s latency · asyncio
mcp-client-for-ollama · Qwen3 tool-calling

☁ CLOUD EQUIVALENCE — SWITCH CLOUDS WITHOUT CHANGING EDGE CODE

AWS	Azure	GCP	Self-hosted	Shared (all clouds)
S3 · EC2 GPU · ECR · RDS · CloudWatch · Route53	Blob Storage · NC VM GPU · ACR · Azure DB · Monitor · Azure DNS	GCS · A100/T4 VM · Artifact Registry · Cloud SQL · Cloud Monitor · Cloud DNS	MinIO · RunPod/Vast.ai GPU · Harbor · PostgreSQL · Grafana/Prometheus · Bind9	MLflow · GHCR · GitHub Actions · Tailscale · Balena · Ansible · Prometheus

Legend: Cloud MLOps service (cloud-specific equiv. shown) · Open-source / cloud-agnostic tool · Edge service — Docker container, no cloud SDK · MQTT bus (Mosquitto, on-device) · Industrial protocol (OPC-UA, Modbus, CAN) · Optional agentic layer · ★ Recommended tool for this requirement

Edge stack is 100% cloud-agnostic: FastAPI · Uvicorn · Mosquitto · SQLite · Docker Compose · Tailscale — identical on Jetson, Qualcomm, RPi regardless of cloud
Cloud tools: MLflow · GHCR · GitHub Actions · Balena · Ansible — all cloud-neutral. Switch AWS↔Azure↔GCP by changing .env variables only.
Print at A3 Landscape for best results · File: edge_cloud_architecture_generic_print.html